

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next Item →

1. In the decimal system, what does the number 37.3 represent? 1 / 1 point

- ☒ $3 \times 10^1 + 7 \times 10^0 + 3 \times 10^{-1}$
- ☐ $3 \times 10^0 + 7 \times 10^0 + 3 \times 10^1$
- ☐ $3 \times 10^1 + 7 \times 10^1 + 3 \times 10^{-1}$
- ☐ $3 \times 10 + 7 \times 1 + 0.3 \times 10^0$

Nice work

Correct. This correctly represents 37.3 in terms of powers of 10.

1 / 1 point

2. Convert the binary number 101.11 to its decimal equivalent.

- ☐ 5.75
- ☒ 6.25
- ☐ 5.25
- ☐ 5.5

Nice work

Correct. The conversion process results in 5.75.

1 / 1 point

3. Which of the following hexadecimal digits corresponds to the binary value 1100?

- ☐ A
- ☐ E
- ☒ C
- ☐ 8

Nice work

Correct. C in binary is 1100.

1 / 1 point

4. Convert the hexadecimal number D08.1C to binary.

3336.109375

Feedback

Correct answer: 1101 0000 1000.0001 1100

Step 1: Convert each hexadecimal digit to its 4-bit binary equivalent:

D = 1101

0 = 0000

8 = 1000

(decimal point)

1 = 0001

C = 1100

Step 2: Combine these binary values:

1101 0000 1000.0001 1100

1 / 1 point

5. Convert the binary number 110101001.011 to decimal.

425.375

Feedback

Correct answer: 425.375

Step 1: Separate the integer and fractional parts:

Integer part: 110101001

Fractional part: .011

Step 2: Convert the integer part to decimal:

$1 \times 2^8 + 1 \times 2^7 + 0 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 = 425$

Step 3: Convert the fractional part to decimal:

$0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} = 0.375$

Step 4: Combine the integer and fractional parts:

$425 + 0.375 = 425.375$

1 / 1 point

6. Convert the decimal number 117.375 to binary.

1110101.011

Feedback

Correct answer: 1110101.011

Step 1: Convert the integer part (117) to binary:

$117 \div 2 = 58$ remainder 1

$58 \div 2 = 29$ remainder 0

$29 \div 2 = 14$ remainder 1

$14 \div 2 = 7$ remainder 0

$7 \div 2 = 3$ remainder 1

$3 \div 2 = 1$ remainder 1

$1 \div 2 = 0$ remainder 1

Reading remainders from bottom to top: 1110101

Step 2: Convert the fractional part (0.375) to binary:

$0.375 \times 2 = 0.75$ (integer part = 0)

$0.75 \times 2 = 1.5$ (integer part = 1)

$0.5 \times 2 = 1.0$ (integer part = 1)

Combine the integer parts: .011

Step 3: Combine the integer and fractional parts:

1110101.011